

# EHTIRAM (EZRA) SHUKUROV

## Software Engineer

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## SUMMARY

Software engineer who builds and ships end to end, from deployed web applications in Svelte and React to a voice-driven AI pipeline on Azure OpenAI and a VR research environment in Unreal Engine 5. Comfortable owning a system from first prototype through the failures that surface in a real build.

## EDUCATION

### University of Minnesota | Minneapolis, MN

Sep 2024 - Dec 2026 (Expected)

M.S., Computer Science | GPA: 3.8/4.0

- Coursework: Developing the Interactive Web, Visualization, Computer Graphics, Intelligent Mixed Reality Systems, Robotics.

### Azerbaijan Technical University | Baku, Azerbaijan

Sep 2020 - Jul 2024

B.S., Information Technology | SABAH Groups Honors Program | GPA: 97/100 (Honors)

## EXPERIENCE

### Software Engineering Intern | IPMD, Inc. (Remote)

Sep 2026 - Present

- Developing EchoSphere Music for IPMD's EchoAI platform: a browser-based tool that generates original instrumental music from a chosen emotional state, plus a Python service that scores video of the EchoSphere robot.
- Working remotely alongside engineers and senior mentors, building with AI-assisted development throughout.

### Graduate Researcher | XR & Perception Lab, UMN

Jan 2026 - Present

- Built a custom Unreal Engine 5 environment with MetaHuman avatars for MS thesis research on avatar realism and stress responses in VR, advised by Prof. Victoria Interrante.
- Resolved GPU shadow-map crashes, texture-baking issues, facial-capture integration problems, and frame jitter; integrated Claude Code with the Unreal Editor through MCP to support debugging.

### Lead Student Worker | UMN Dining Services

Oct 2024 - May 2026

- Trained student workers, coordinated shift coverage, and communicated operational issues to supervisors.

### Customer Service Associate | International Bank of Azerbaijan (ABB)

Apr 2024 - Jul 2024

- Resolved customer application, account, and payment issues through daily direct support.

### Unity Developer Intern | Azerbaijan Technical University

Jul 2022 - Oct 2022

- Developed game mechanics and interactive 3D prototypes in Unity and C#, taking the demo game Sharky from concept to a playable build.

## SELECTED PROJECTS

### Dimension | SvelteKit, D3.js, TypeScript

[Live Demo](#)

- Integrated five team-built visualizations into a cohesive scrollytelling app exploring 100,000 movie ratings; unified frontend design, fixed SVG scaling bugs, and implemented sticky filters and animations.
- Developed an interactive age/genre line chart with genre toggles, sample-size indicators, tooltips, and scroll-driven transitions; refined labels and interactions using peer feedback.

### Mixed Reality Thrombectomy Simulator | Unity, C#, Meta Quest 3

[GitHub](#)

- Replaced physics-based catheter navigation with a 917-node directed graph derived from 3,226 vascular centerline points, supporting seven bifurcations and exact path retraction via a history stack; built haptic feedback and fluoroscopy mode.

### Mixed Reality AI Assistant | Unity, C#, Azure OpenAI

[GitHub](#)

- Built a speech-to-speech pipeline with Azure OpenAI and Azure Speech REST APIs, combining function calling and live camera context for scene-aware voice interaction on Meta Quest 3.

### Summer Movies | SvelteKit, D3.js, Three.js

[Live Demo](#)

- Created interactive visualizations of 899 films, including bar, bump, and heatmap charts, a brushable timeline, and a Three.js/WebGL scene with scroll-synchronized camera movement.

### Platform Ink | C++, OpenGL, GLSL, SDL3

[GitHub](#)

- Built a third-person 3D platformer from scratch with a custom rendering pipeline featuring toon shading, rim lighting, and inverted-hull outlines.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, C#, C++, SQL, HTML/CSS

**Web:** Svelte/SvelteKit, React, D3.js, Three.js/WebGL, Tailwind CSS

**Backend & AI:** ASP.NET, REST APIs, Azure OpenAI, Azure Speech, function calling

**Graphics & XR:** Unreal Engine 5, Unity, OpenGL, HLSL/GLSL, MetaHuman, Meta Quest SDK, Blender

**Tools:** Git, GitHub, Vercel, Claude Code, MCP, debugging and profiling

**Spoken languages:** English (Fluent; IELTS 8.0), Azerbaijani (Native), Turkish (Proficient)